SN6200-LD Engineering Sample

Release Notes

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SN6200-LD Engineering Sample

This document provides information about the SN6200-LD engineering sample for Cumulus Linux 5.18.

Important:

- The SN6200-LD engineering sample requires Cumulus Linux release 5.18.6200. For information about Cumulus Linux 5.18, refer to the Cumulus Linux 5.18 User Guide.
- Future upgrades from the SN6200-LD engineering sample will be binary upgrades using ONIE.

Supported Platforms

The SN6200-LD engineering sample supports the following platforms:

Model Number	Part Number	Description
SN6200-LD	920-9N60F-00LI-GC0	NVIDIA Spectrum-6 based 800GbE Ethernet switch with Cumulus Linux Authentication, Rack Scale, 32xOSFP, 1xQSFP112, 256x200G Cable Cartridge, 48VDC Busbar, 8-Cores CPU, Secure-boot, Liquid-Cooling, Rail Kit

SN6200-LD Supported Port Speeds

Speed [GbE]	Auto-Negotiation	Force			Cables/Modules	Cable Length [m]
		RS FEC	FC FEC	NO FEC		
200G PAM4: 800G 400G 200G	✓	✓	✗	✗	Optic	Up to 100
					Copper	1
100G (service port)	✓	✓	✗	✗	Optic	Up to 100

- PAM4 is the system default.
- When connecting NVIDIA-to-NVIDIA in PAM4, make sure Auto-negotiation is enabled.
- 100 Gbps serdes lane speed is not supported

Package Content

NVIDIA has validated this release with the following component versions:

Package	Version
SDK	4.10.1526-002
Kernel	6.1.123-1
MFT	4.36.0.1037
Hardware management	7.0070.1133
NetQ	5.2.1
FRR	10.0.3
NVUE Core	1.14.0.77
UMF	91.1.14.14
BMC	88.0060.2422

New Features

-

Unsupported Features

The Spectrum-6 engineering sample does not support:

- Optimized image upgrade
- Package upgrade
- ISSU upgrade
- Fast reboot
- Precision Time Protocol (PTP)

Network Port Connectivity Matrix

Supported Speeds	Connectivity	Cable PN	Cable Type and Description	Lengths (m)
800G_R4 to 200G_R1	SN6200-LD ↔ SN6200-LD/ ConnectX-9	980-9IAU1-00X M00	DR4 Optic module, OSFP flat	Up to 100
800G_R4 to 200G_R1	SN6200-LD ↔ SN6200-LD	980-9IAM6-00X 001	ACC copper, OSFP flat	1, 2, 3

When connecting to ConnectX-9 using cable PN 980-9IAU1-00XM00, ensure that the adapter is running firmware version 82.49.10xx or later and apply the configuration setting `mlxconfig -d /dev/mst/mt4133_pciconf0 set MODULE_MAX_POWER_OVERRIDE_M0=0x96.`

Service Port Connectivity Matrix

Supported Speeds	Connectivity	Cable PN	Cable Type and Description	Lengths (m)
100G_R1	SN6200-LD	980-9I068-00NM00 (MMS1X00-NS400)	DR4 Optic module, QSFP112 flat	Up to 100

Verified Cables

Speed	Part Number	Description
1600GbE	980-9IAU1-00XM00	NVIDIA twin port transceiver, 2xDR4, 1600Gbps, OSFP, 2xMPO(APC), 1310nm SMF, up to 500m, RHS, FRO, Gen2, Ethernet Only
1600GbE	980-9IAM6-00X001	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 1.1m, RHS to RHS
1600GbE	980-9IAM6-00X002	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 2m, RHS to RHS
1600GbE	980-9IAM6-00X003	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 3m, RHS to RHS

NVIDIA DSX Air

NVIDIA DSX Air only supports 256 ports (the physical switch supports 512 ports).

Install BMC

The Spectrum-6 switch requires you to download and install the Baseboard Management Controller (BMC) onto the switch.

BMC is a specialized microcontroller designed to deliver out-of-band remote monitoring and management for servers and switches. Operating independently from the main CPU and operating system, the BMC enables administrative control even when the switch is powered down or unresponsive. It streamlines server and switch management while enhancing network efficiency, reliability, and security through automation of critical tasks.

For information about downloading and installing BMC, refer to the BMC section of the Cumulus Linux 5.18 user guide

Liquid Cooling Leakage Telemetry

Liquid-cooled NVIDIA switches contain leakage sensors that detect coolant leaks. These sensors are critical for data center safety as an undetected leak can cause equipment damage, fire risk, or unplanned downtime.

This feature adds leakage sensor data to the gNMI and OTEL telemetry pipeline, enabling remote monitoring systems, such as Prometheus, Grafana, and gNMI collectors, to subscribe to real-time leakage status and trigger automated alerts.

Open Issues

ID	Description

Legal Notices and 3rd Party Licenses

Product	Version	Legal Notices and 3rd Party Licenses
SDK	4.10.0290	> 3rd Party Unify Notice > 3rd Party Notice > License
MFT	4.35.0-2064	> 3rd Party Notice—Linux > License—Linux

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